Abstract

Warehouses often rely on manual processes to monitor shelf inventory, leading to errors, delays, and increased workload for staff.

The lack of real-time visibility results in inefficient operations, poor inventory management, and higher operational costs.Further more the warehouse experience loss in profit due to time delays caused by the traditional way of warehouse management. Traditional processes cause the creation of unsafe working conditions for the workers that usually end up in accidents.

This project proposes a smart warehouse monitoring system that uses ultrasonic sensors and a microcontroller to detect the presence or removal of containers on shelves. The system continuously measures distance to identify changes in shelf occupancy and automatically updates inventory data in real time. By reducing the need for manual inspection, the proposed solution improves accuracy, efficiency, and overall warehouse management while minimizing human effort.

CHAPTER 1

INTRODUCTION

1.1 Purpose of the Report

The purpose of this report is to present the design and development of a smart warehouse monitoring system that automates the detection of container presence on storage shelves. The system aims to reduce manual effort, improve inventory accuracy, and enable real-time tracking using sensor-based technology. This report also demonstrates how design thinking is applied to identify user problems and develop an efficient, low-cost solution.

1.2 Overview of the Design Thinking Approach

Design Thinking is a user-centric approach used to solve real-world problems by focusing on the needs and challenges of users. It involves five key stages: **Empathize, Define, Ideate, Prototype, and Test**. In this project, the approach begins with understanding the difficulties faced by warehouse staff and managers, followed by clearly defining the problem of inefficient manual inventory tracking. Various solutions are explored, leading to the development of a sensor-based prototype that automates shelf monitoring. The final system is evaluated based on its effectiveness in improving accuracy and reducing manual effort.

1.3 Problem Statement / Challenge

Warehouses rely heavily on manual processes to monitor shelf inventory, which leads to errors, delays, and increased workload for workers. Continuous physical inspection of shelves is time-consuming and inefficient, while the lack of real-time data affects decision-making for managers.

This challenge highlights the need for an automated system that can detect the presence or removal of containers on shelves and update inventory information in real time, improving overall efficiency and reducing operational costs.

CHAPTER 2: EMPATHISE (Understanding the User)

2.1 Research Objectives

The objective of this study is to understand the challenges faced by warehouse workers and managers in monitoring shelf inventory. The research focuses on identifying issues related to manual stock checking, accuracy of inventory data, and the impact of these challenges on efficiency and workload. It also aims to explore user needs for a more automated and reliable inventory monitoring system.

2.2 Research Methods

We gathered insights in the following way:

1. Observation: Studied how inventory is manually checked in warehouse-like environments or storage setups.
2. Interviews: Informal discussions with individuals familiar with warehouse operations or stock management.
3. Surveys: Basic questionnaires to understand common issues such as time consumption, errors, and workload.

2.3 Key Insights from User Research

The research revealed several important insights:

Pain Points

- Manual checking of shelves is time-consuming and repetitive
- High chances of human error in inventory tracking
- Requires continuous movement, causing physical fatigue
- Lack of real-time updates leads to delays

Needs

- A system for automatic detection of container presence
- Real-time inventory updates
- Reduction in manual effort and physical workload

Desires

- Faster and more efficient warehouse operations
- Improved accuracy in stock management
- A simple and cost-effective automation solution

2.4 User Persona

- Persona 1: Warehouse Worker
- Works long hours managing stock
- Regularly checks shelves manually
- Faces fatigue due to repetitive tasks
- Needs an easier and faster way to monitor inventory

- Persona 2: Warehouse Manager
- Responsible for inventory accuracy
- Relies on reports and manual updates
- Faces issues with delayed or incorrect data
- Needs real-time visibility and better control

CHAPTER 3

STAGE 2 – DEFINE (FRAMING THE PROBLEM)

3.1 Root Cause Analysis (5 Why Analysis)

The inefficiency in warehouse inventory monitoring can be understood using the 5 Why Analysis:

Problem: Inventory is not updated accurately in real time.

1. Why?

   Because workers manually check shelf status. By the time it gets updated the numbers are out of sync.

2. Why?

   Because there is no automated system to detect container presence.

3. Why?

   Because existing systems focus more on environmental monitoring rather than shelf occupancy.

4. Why?

   Because implementing advanced automation systems can be costly and complex.

5. Why?

   Because warehouses often lack low-cost, simple solutions for real-time inventory tracking.

Root Cause:

Lack of a simple, cost-effective automated system to detect and update shelf-level inventory in real time.

3.2 Problem Definition (How Might We Questions)

Based on the identified challenges, the problem can be reframed as:

1. How might we automate the detection of container presence on warehouse shelves?
2. How might we reduce manual effort and errors in inventory monitoring?
3. How might we provide real-time updates of shelf status to warehouse managers?
4. How might we design a low-cost and scalable solution for warehouse automation?

3.3 Synthesized Problem Statement

Warehouse inventory monitoring is currently dependent on manual processes, leading to inefficiencies, errors, and delays in updating stock information. There is a need for a cost-effective and automated system that can detect the presence or removal of containers on shelves and provide real-time inventory updates, thereby improving operational efficiency and reducing human effort.

3.4 Scope of the Project

The scope of this project includes the design and development of a prototype system that:

- Uses ultrasonic sensors to detect the presence or absence of containers on shelves
- Utilizes a microcontroller (ESP32) to process sensor data
- Provides real-time updates of inventory status
- Demonstrates automation for a small number of shelves (prototype level)
- Integrates with a software interface (such as Excel or dashboard) for monitoring

Chapter 4

Stage 3 – Ideate (Generating Ideas)

4.1 Brainstorming Techniques Used

To generate innovative solutions for warehouse automation, brainstorming sessions were conducted using structured techniques such as mind mapping and group discussions. The team explored multiple approaches by analyzing the problem from different perspectives, including cost, efficiency, scalability, and ease of implementation.

Techniques like "How Might We" questions and rapid idea listing were used to encourage creative thinking without limitations.

4.2 Ideas Generated

Several ideas were proposed during the ideation phase to address the problem of manual warehouse monitoring. These included:

- RFID-based tracking system for containers
- Camera-based monitoring using computer vision
- Weight sensors on shelves to detect load changes
- Ultrasonic sensor-based system to measure distance and detect container presence
- Infrared (IR) sensors for object detection

Each idea aimed to automate inventory tracking and reduce human intervention while improving accuracy and efficiency.

4.3 Selection Criteria

The generated ideas were evaluated based on specific criteria such as cost-effectiveness, ease of implementation, accuracy, scalability, and suitability for a college-level prototype. Systems like RFID and computer vision were found to be complex and expensive for implementation. The ultrasonic sensor-based approach was selected as it is low-cost, simple to implement, and effective in detecting the presence or absence of containers through distance measurement.

4.4 Final Idea Selection

Based on the evaluation, the ultrasonic sensor-based warehouse monitoring system was chosen for further development. This solution meets the project requirements by providing real-time detection of container presence while remaining affordable and easy to implement within the given constraints.

CHAPTER 5

STAGE 4 – PROTOTYPE (CREATING SOLUTIONS)

5.1 Description of Prototype Developed

The prototype developed in this project is a smart warehouse shelf monitoring system designed to detect the presence or removal of containers using ultrasonic sensors. Each sensor is mounted above a designated shelf position and continuously measures the distance between the sensor and the object placed below it. When a container is present, the measured distance remains within a predefined range, whereas the removal of the container results in an increase in distance.

The collected sensor data is processed using a microcontroller, which determines the status of each shelf as either occupied or empty. This status is then transmitted to a connected system, where it is displayed and recorded for monitoring purposes. The prototype demonstrates real-time inventory tracking on a small-scale warehouse setup.

5.2 Tools, Materials, and Software Used

Hardware Components

- Ultrasonic sensors (HC-SR04) for distance measurement
- ESP32 microcontroller for processing and wireless communication
- Breadboard and jumper wires for circuit connections
- Power supply (USB or adapter)
- Physical shelf model (cardboard or wooden structure)

Software Tools

- Arduino IDE for programming the ESP32
- Embedded C/C++ for implementing sensor logic
- Python (or equivalent backend) for data handling
- Excel / Google Sheets for displaying and recording inventory status

5.3 Working of the Prototype

The system operates based on continuous distance measurement and comparison with predefined thresholds:

- The ultrasonic sensor emits sound waves and measures the time taken for the echo to return.
- The ESP32 calculates the distance using the received signal.
- The measured distance is compared with a predefined threshold value.
- If the distance is within the threshold range, the container is considered present.
- If the distance exceeds the threshold, the container is considered removed.
- The updated status is transmitted to the connected system for real-time monitoring.

This process is repeated continuously to ensure accurate and up-to-date inventory tracking.

5.4 Alignment with User Needs

The prototype effectively addresses the user needs identified during the earlier stages of the design thinking process:

- Reduction in manual effort: Eliminates the need for continuous physical inspection of shelves
- Improved accuracy: Reduces human errors in inventory monitoring
- Real-time visibility: Provides instant updates on shelf status to warehouse managers
- Operational efficiency: Enhances speed and effectiveness of warehouse management

By emphasizing simplicity, affordability, and ease of implementation, the proposed prototype offers a practical solution that can be scaled for real-world warehouse applications.

Chapter 6: Stage 5 – Test (Validating Solutions)

6.1 User Testing Process and Methods

The developed warehouse monitoring system was tested in a controlled prototype environment consisting of multiple shelves equipped with ultrasonic sensors connected to a microcontroller (ESP32). The system was evaluated by simulating real warehouse scenarios, such as placing and removing containers from shelves.

Testing was conducted with a small group of users, including students and individuals acting as warehouse operators. The testing process focused on verifying the accuracy of container detection, response time of the system, and real-time updating of inventory data in the connected Excel sheet.

6.2 Feedback Received from Users

Users observed that the system was effective in detecting the presence and removal of containers in real time. The automatic updating feature reduced the need for manual checking and was appreciated for its simplicity and efficiency.

However, some users noted minor issues such as occasional fluctuations in sensor readings and the need for proper sensor placement to ensure accurate detection. Suggestions were also made to improve the user interface for better visualization of shelf status.

6.3 Iterations Made Based on Feedback

Based on the feedback received, several improvements were implemented in the system. Sensor readings were stabilized by averaging multiple readings to reduce noise and improve accuracy. The placement of sensors was adjusted to ensure consistent and reliable measurements.

Additionally, the system logic was refined by introducing threshold calibration to better distinguish between the presence and absence of containers. Minor improvements were also made to the data display format in the Excel sheet for clearer understanding and usability.

Here's a clean, well-structured Chapter 7 – Conclusion (Reflections & Learnings) you can directly use:

CHAPTER 7

CONCLUSION (REFLECTIONS & LEARNINGS)

7.1 Key Takeaways from the Process

This project demonstrated how the Design Thinking approach can be effectively applied to solve real-world problems in warehouse management. By focusing on user needs, the project identified inefficiencies in manual inventory monitoring and proposed a practical, sensor-based solution. The development of the prototype provided hands-on experience in integrating hardware and software components, including sensors, microcontrollers, and data handling systems.

A key takeaway from this process is the importance of simplicity and cost-effectiveness in designing solutions that can be realistically implemented. The project also highlighted how real-time data can significantly improve operational efficiency and decision-making.

7.2 What Worked Well and What Didn't

What worked well:

The ultrasonic sensor-based detection system successfully identified the presence and absence of containers.

The integration of the ESP32 with software tools enabled real-time data transmission and monitoring.

The overall system was simple, low-cost, and easy to implement, making it suitable for prototype-level demonstration.

What didn’t work well:

Sensor readings were sometimes inconsistent due to environmental factors such as surface irregularities or object positioning.

The system was limited to small-scale implementation and may require optimization for larger setups.

Direct integration with Excel or software required additional configuration and handling.

7.3 Opportunities for Improvement

The current prototype can be enhanced in several ways to improve performance and scalability:

Using more precise sensors such as Time-of-Flight sensors for improved accuracy

Implementing a centralized database instead of direct Excel updates for better scalability

Adding a graphical dashboard for better visualization of inventory status

Expanding the system to handle a larger number of shelves and sensors

Integrating additional technologies such as RFID or computer vision for advanced automation

These improvements can help transform the prototype into a more robust and industry-ready solution.